TMUA / PRACTICE PAPER

Set C Paper 2

Mathematical Reasoning 20 questions · 75 minutes

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Trigonometry

1. A triangle ABC has $\angle BAC = 30°$, $BC = 2$ and $AC = 2\sqrt{3}$.

Two different triangles satisfy these conditions.

What is the sum of their areas?

A $\sqrt{3}$

B $2\sqrt{3}$

C $3\sqrt{3}$

D $4\sqrt{3}$

E $6\sqrt{3}$

Logic

2. Let $f(x)$, $g(x)$ and $h(x)$ be real functions and consider the statement

$$\text{If } x > 0 \text{ and } f(x) \leq 0 \text{ then } g(x) \neq 1 \text{ or } h(x) \geq -1 \qquad (*)$$

Which of the following statements is the *contrapositive* of $(*)$?

A If $g(x) = 1$ and $h(x) < -1$ then $x \leq 0$ or $f(x) > 0$

B If $g(x) = 1$ or $h(x) < -1$ then $x \leq 0$ or $f(x) > 0$

C If $x \leq 0$ or $f(x) > 0$ then $g(x) = 1$ and $h(x) < -1$

D If $g(x) \neq 1$ or $h(x) \geq -1$ then $x > 0$ and $f(x) \leq 0$

E If $x > 0$ and $f(x) \leq 0$ then $g(x) = 1$ and $h(x) < -1$

F If $g(x) = 1$ and $h(x) < -1$ then $x \leq 0$ and $f(x) > 0$

Logic

3. Consider the equation

$$4^x - 3 \times 2^x + k = 0$$

where k is a real constant.

Which of the following statements is/are true?

I For the equation to have a real solution, it is necessary that $k < 0$

II For the equation to have a real solution, it is sufficient that $k < 0$

III For the equation to have two distinct real solutions, it is necessary and sufficient that $0 < k < \frac{9}{4}$

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Logic

4. An infinite geometric series

$$a + ar + ar^2 + ar^3 + \cdots$$

where $a \neq 0$, has sum to infinity S.

Which of the following statements is/are true?

I $S > a$ **only if** $r > 0$

II $r > 0$ is **sufficient** for $S > a$

III a and r being both positive or both negative is **necessary and sufficient** for $S > a$

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Proof & Error Analysis

5. A student proves that $\sqrt{2}$ is irrational, using proof by contradiction. The proof uses the following five statements, presented here in a scrambled order.

(1) Since p is even, write $p = 2k$ for some positive integer k; substituting into $p^2 = 2q^2$ gives $4k^2 = 2q^2$, so $q^2 = 2k^2$, and therefore q is even.
(2) Suppose, for contradiction, that $\sqrt{2}$ is rational, so $\sqrt{2} = \dfrac{p}{q}$ for some positive integers p and q with no common factor greater than 1.
(3) This means p and q share the common factor 2, contradicting the assumption that p and q have no common factor greater than 1.
(4) Squaring both sides of $\sqrt{2} = \dfrac{p}{q}$ gives $p^2 = 2q^2$, so p^2 is even, and therefore p is even.
(5) Therefore the original assumption is false, and $\sqrt{2}$ is irrational.

Which ordering of the statements (1)–(5) gives a valid proof, with each statement following logically from those before it?

A 2, 4, 1, 3, 5

B 2, 1, 4, 3, 5

C 4, 2, 1, 3, 5

D 2, 4, 3, 1, 5

E 2, 4, 1, 5, 3

F 2, 3, 4, 1, 5

G 2, 4, 3, 5, 1

H No ordering of the statements gives a valid proof.

Sequences & Series

6. A geometric progression has a real first term and a real common ratio.
The sum of the first three terms is 14, and the sum of the first six terms is 126.
What is the sum of the first nine terms?

A 98

B 252

C 511

D 1022

E 1092

F 1134

Logic

7. Let θ be a real number with $0 < \theta < 2\pi$.

Consider the following three statements about the inequality $\sin\theta + \cos\theta > 1$

I: A necessary condition for the inequality to hold is that $\sin\theta\cos\theta > 0$

II: A sufficient condition for the inequality to hold is that $\sin\theta\cos\theta > 0$

III: A necessary and sufficient condition for the inequality to hold is that $\sin\theta\cos\theta > 0$ and $\sin\theta + \cos\theta > 0$

Which of the statements is/are true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

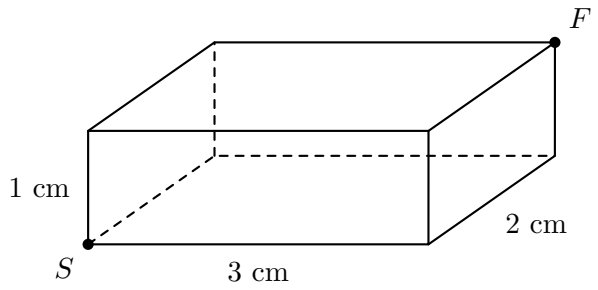
Geometry

8. A solid cuboid has edges of length 1 cm, 2 cm and 3 cm.

An ant starts at one corner S of the cuboid and crawls across the outside surfaces to the corner F furthest from it. That is, the corner not sharing any face with the starting corner.

The ant may cross any face but cannot pass through the inside of the cuboid.

What is the shortest possible distance the ant travels?



A $\sqrt{14}$ cm

B $3\sqrt{2}$ cm

C $2\sqrt{5}$ cm

D $1 + \sqrt{13}$ cm

E $\sqrt{26}$ cm

F 6 cm

Integration

9. Let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x.

Define

$$f(x) = \lfloor 2x \rfloor - \lfloor x \rfloor$$

for all real x.

The value of $\displaystyle\int_0^2 f(x)\,\mathrm{d}x$ is

A 0

B $\frac{1}{2}$

C 1

D $\frac{3}{2}$

E 2

F 3

G 4

Logic

10. Let f be a polynomial defined for all real numbers x.

Let g be the function defined by

$$g(x) = \int_0^x f(t)\,\mathrm{d}t$$

for all real numbers x.

Consider the following three conditions on f:

1 $xf(x) \geq 0$ for all real numbers x

2 $g(x) \geq 0$ for all real numbers x

3 $f(0) = 0$

Consider also the following three statements:

I Condition 1 is necessary for Condition 2

II Condition 1 is sufficient for Condition 2

III Condition 3 is necessary for Condition 2

Which of the statements I, II and III is/are true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Number

11. Let $N = 2^8 \times 3^6$

A positive integer k is called *suitable* if both $\frac{N}{k^2}$ and $\frac{k^3}{N}$ are integers.

What is the sum of all suitable values of k?

A 432

B 576

C 648

D 864

E 1296

Proof & Error Analysis

12. In this question you may use the facts

$$\sin\frac{\pi}{12} = \frac{\sqrt{6}-\sqrt{2}}{4} \qquad \text{and} \qquad \sin\frac{5\pi}{12} = \frac{\sqrt{6}+\sqrt{2}}{4}$$

A student attempts to answer the following question.

Find all real values of x, with $0 < x < \pi$, such that
$\log_2(\sin x) + \log_2(\cos x) = -2$.

The student's attempt is as follows.

I Using the laws of logarithms, the equation becomes $\log_2(\sin x \cos x) = -2$

II Hence $\sin x \cos x = \dfrac{1}{4}$

III Every solution must therefore satisfy $\sin^2 x \cos^2 x = \dfrac{1}{16}$

IV Writing $\cos^2 x = 1 - \sin^2 x$ and $s = \sin^2 x$ this becomes $16s^2 - 16s + 1 = 0$

V Hence every solution must have $s = \dfrac{2 \pm \sqrt{3}}{4}$, and since $\sin x > 0$ for $0 < x < \pi$, $\sin x = \dfrac{\sqrt{6} \pm \sqrt{2}}{4}$

VI Therefore $x = \dfrac{\pi}{12}, \dfrac{5\pi}{12}, \dfrac{7\pi}{12}, \dfrac{11\pi}{12}$

Which of the following best describes the attempt?

A The attempt is completely correct.

B The attempt is incorrect, and the first error occurs in line I.

C The attempt is incorrect, and the first error occurs in line III.

D The attempt is incorrect, and the first error occurs in line IV.

E The attempt is incorrect, and the first error occurs in line V.

F The attempt is incorrect, and the first error occurs in line VI.

Number

13. There is a non-negative integer k such that the binomial coefficient $\binom{64}{32}$ is divisible by 2^k but not by 2^{k+1}

What is the value of k?

A 0

B 1

C 2

D 16

E 31

F 32

G 63

Probability & Counting

14. Two cards are drawn in order, without replacement, from cards numbered 1 to 6.

Consider the events

A The first card has an even number

B The second card has an even number

C The sum of the two numbers is odd

We say that two events are *independent* if the probability that both occur is equal to their individual probabilities multiplied together.

In notation,

$$A \text{ and } B \text{ are independent} \quad \iff \quad P(A \text{ and } B) = P(A) \times P(B)$$

Which pairs of the events A, B and C are independent?

A None of the three pairs

B A and B only

C A and C only

D B and C only

E A and C; B and C

Probability & Counting

15. A positive integer is called *ascending* if it has at least two digits and, reading its digits from left to right, each digit is strictly greater than the one before it.

For example, 259 is ascending, but 2556 and 92 are not.

How many ascending integers are there less than 10000?

A 126

B 210

C 246

D 255

E 375

F 495

G 3600

Number

16. Let n be a positive integer. Consider the following conditions:

I: n is even
II: n is a multiple of 3
III: n is a prime greater than 3

Which of these conditions is/are **sufficient** for the statement:

$$2^n - 1 \text{ is divisible by } 7$$

A I only

B II only

C III only

D I and II only

E I and III only

F II and III only

G I, II, and III

H None of them

Proof & Error Analysis

17. A statement is given:

$$\text{S: For all real numbers } x, \textbf{ if } x^2 \geq 1 \textbf{ then } x^4 \geq x^2$$

A student wishes to prove S by contradiction and writes the following argument:
Assume that there exists a real number x such that $x^2 \geq 1$ **and** $x^4 < x^2$

I: Then $x^4 - x^2 < 0$

II: So $x^2(x^2 - 1) < 0$

III: So $x^2 - 1 < 0$

IV: So $x^2 < 1$, which contradicts $x^2 \geq 1$

Therefore the original statement S is concluded to be true.
Which of the following best describes this argument?

A It is completely correct.

B It is incorrect, but it would be correct if written in the reverse order.

C It is incorrect, and the student has actually shown that S is false.

D It is incorrect because line **II** does not follow from line **I**.

E It is incorrect because line **III** does not follow from line **II**.

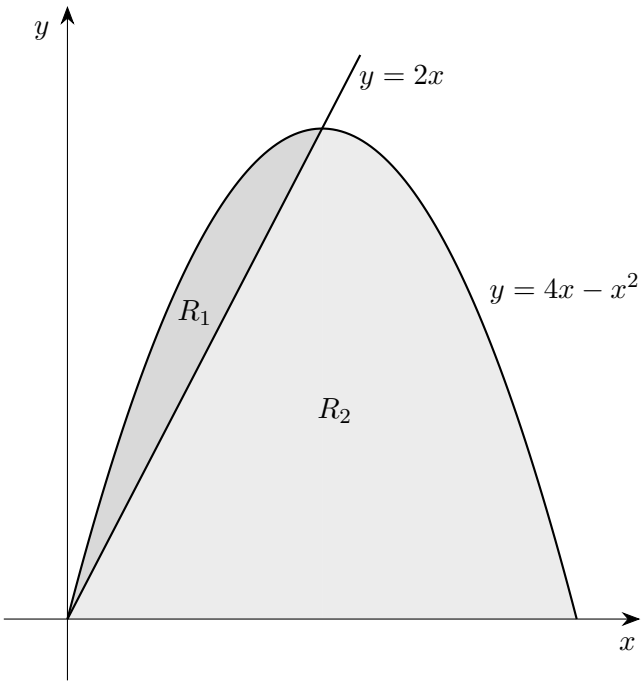
F It is incorrect because the student has not really used proof by contradiction.

Integration

18. The region R in the first quadrant is bounded by the curve $y = 4x - x^2$ and the x-axis.

The line $y = 2x$ divides R into two sub-regions: R_1, which lies above the line, and R_2, which lies below the line.

What is the ratio of the area of R_1 to the area of R_2?



A 1 : 3

B 1 : 4

C 1 : 7

D 1 : 8

E 1 : 15

Exponentials & Logarithms

19. The curve C has equation

$$\log_2(2y+4) = 2\log_2(x+1)$$

For which values of the real constant m does the line $y = mx$ meet C at exactly two distinct points?

A $m < 2$

B $m \leq 2$

C $m > 0$

D $m > 2$

E $m \geq 2$

Logic

20. For a real constant a, the function f is defined for $x > 0$ by $f(x) = \sqrt{x} + \dfrac{a}{\sqrt{x}}$

Consider the conditions

I $f'(4) > 0$

II $f(4) > f(1)$

Which of the following describes the relationship between conditions I and II?

A I is necessary but not sufficient for II.

B I is sufficient but not necessary for II.

C I is both necessary and sufficient for II.

D I is neither necessary nor sufficient for II.

E I and II cannot both be true.